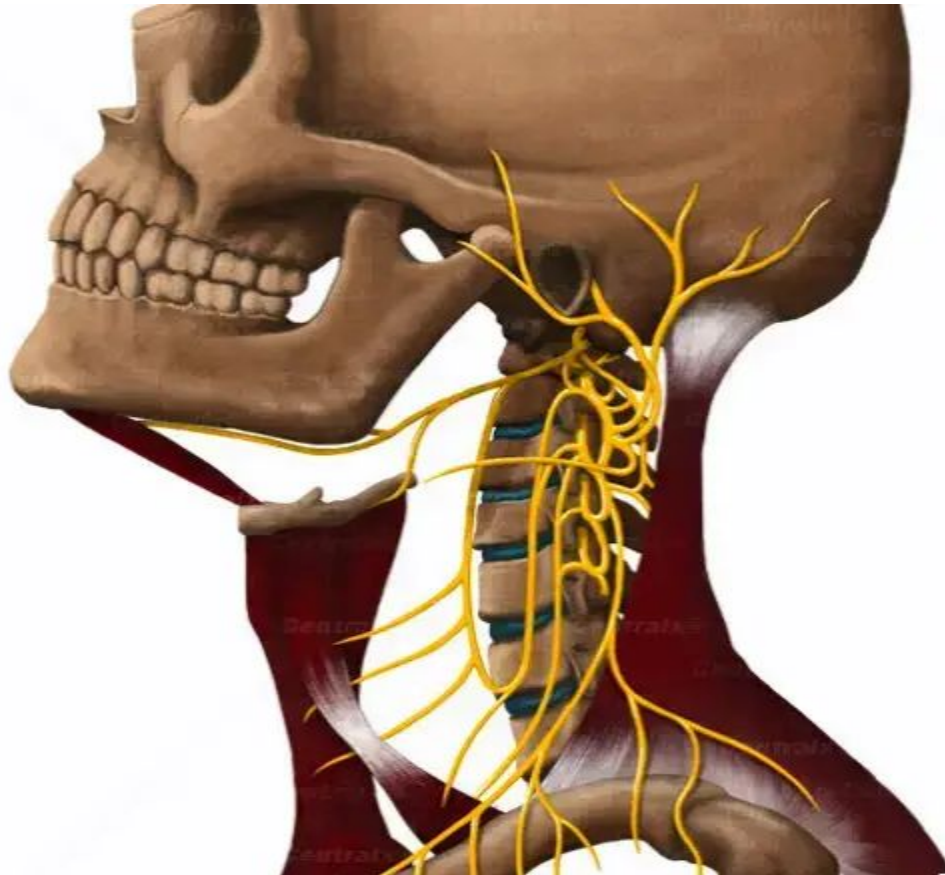


The CERVICAL PLEXUS



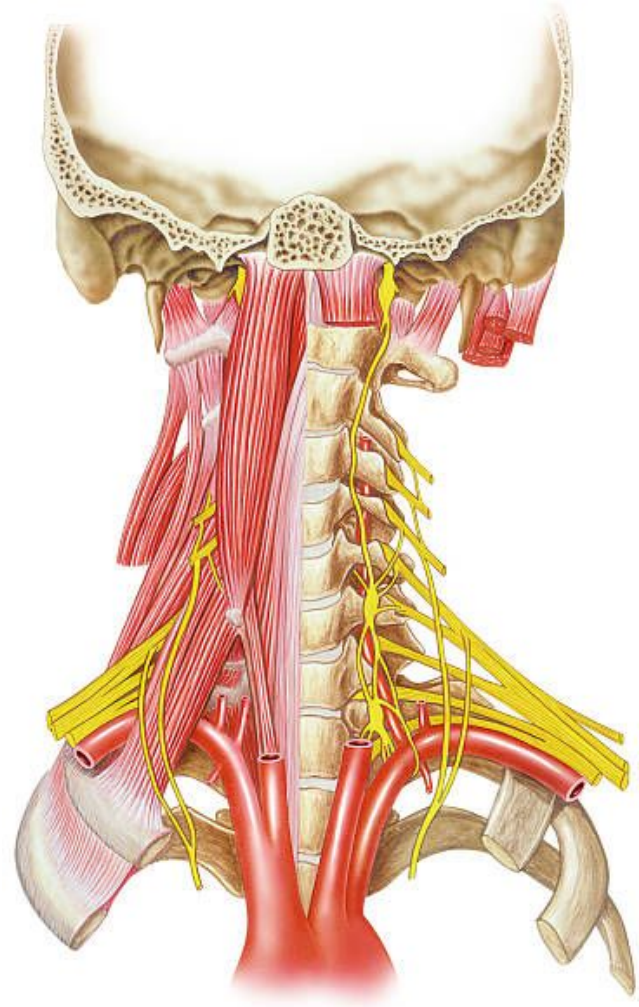
Learning Objectives:

At the end of the lecture, the students should be able to:

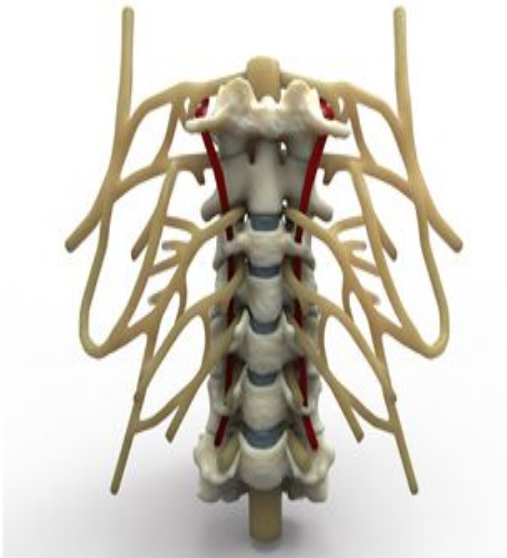
- **Define a peripheral plexus of nerves and its significance.**
- **Describe the formation of the cervical plexus.**
- **Enlist the branches of the cervical plexus and their distribution.**
- **Define the ansa cervicalis, its branches and their distribution.**

Peripheral Nerve Plexuses:

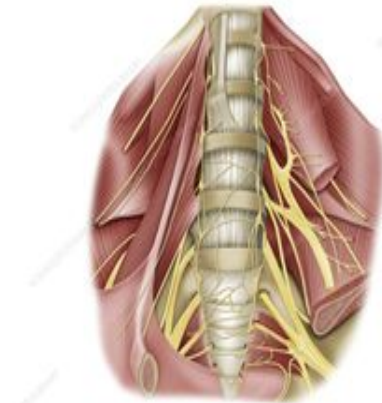
- Peripheral nerves are composed of bundles of nerve fibers.
- They sometimes divide into branches to join neighboring peripheral nerves.
- If this phenomenon occurs frequently, a network of nerves called a **nerve plexus** is formed.



The Cervical Plexus:

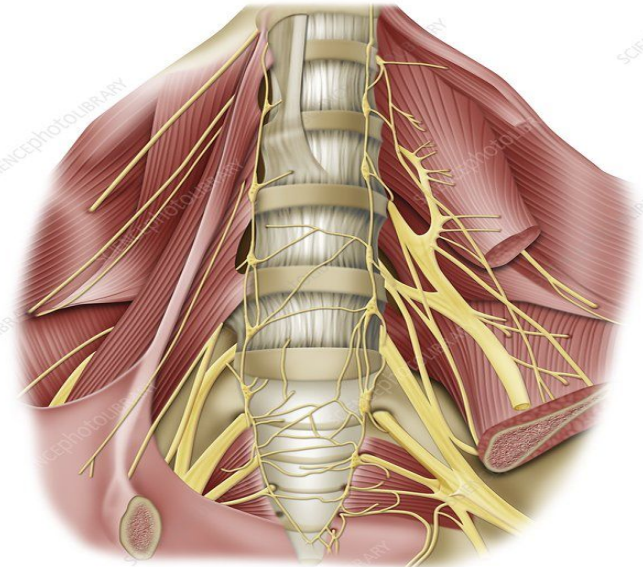
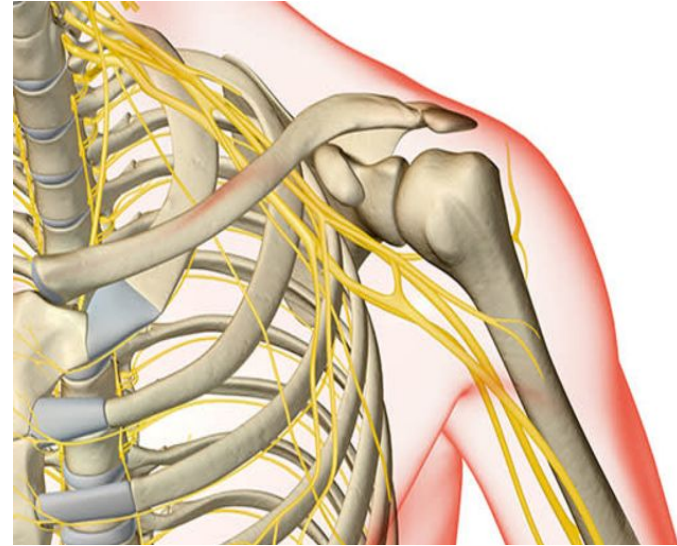


The lumbar & Sacral plexuses:



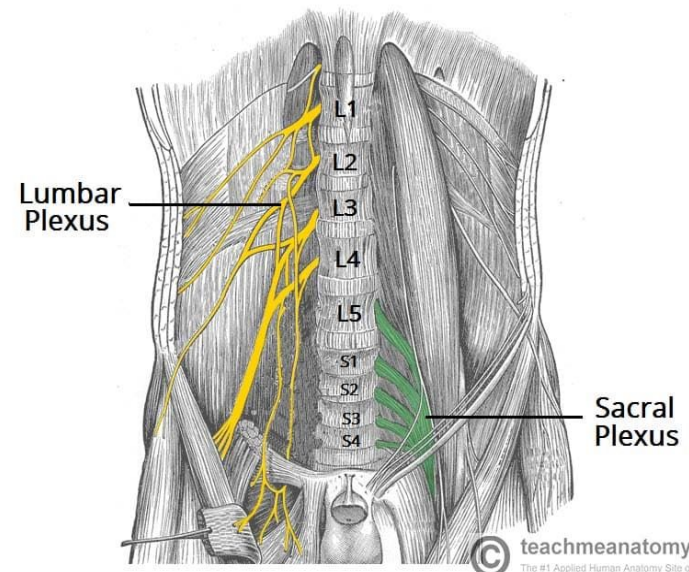
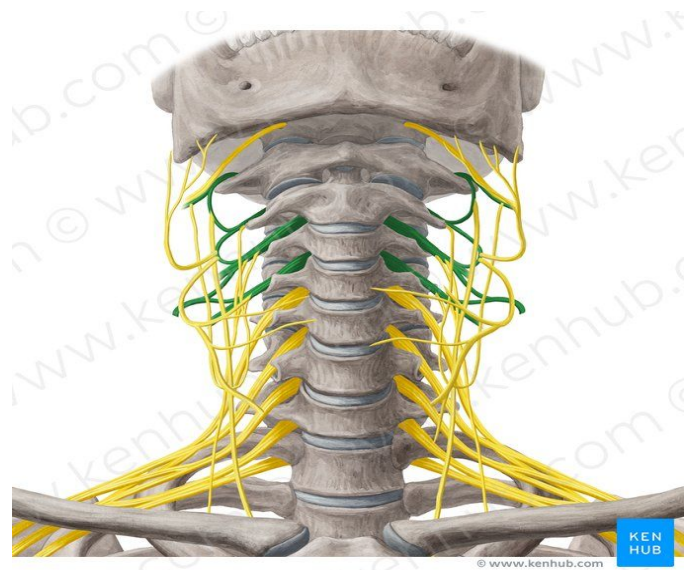
Significance of a Nerve Plexus:

- The formation of a nerve plexus allows individual nerve fibers to pass from one peripheral nerve to another.
- A plexus thus permits a redistribution of nerve fibers within different peripheral nerves.



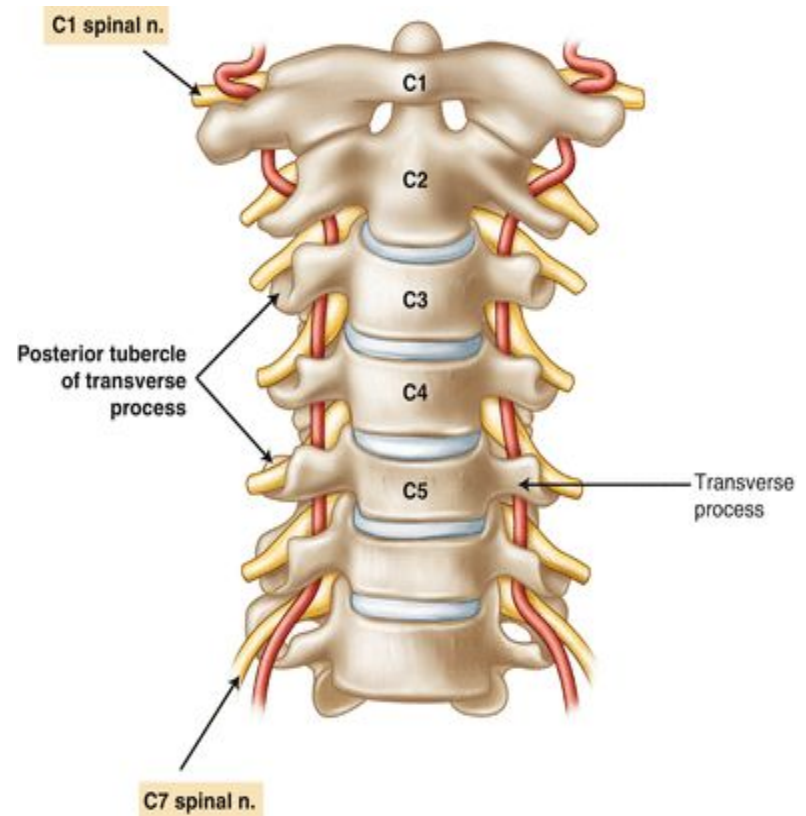
Examples of Nerve Plexuses:

1. The Cervical & Brachial Plexuses are located at the root of the upper limbs, and
2. The Lumbar & Sacral Plexuses are located at the root of the lower limbs.



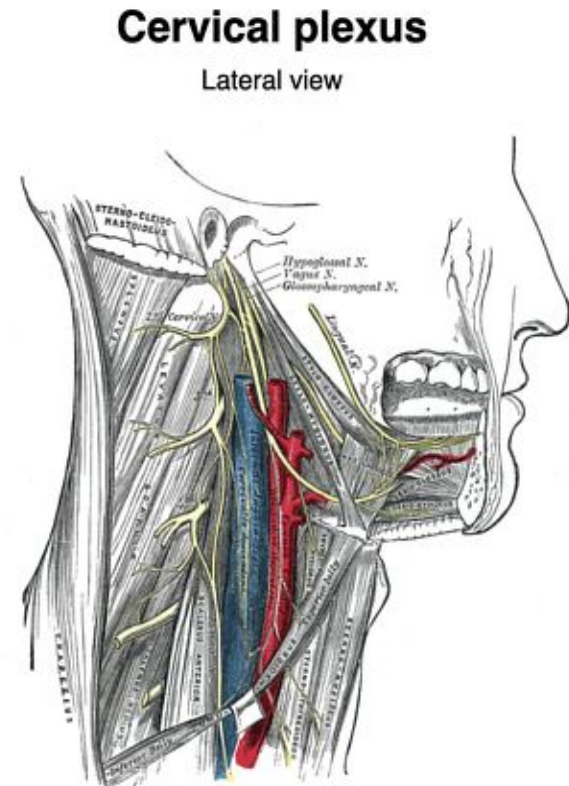
Formation of the Cervical Plexus:

- It is formed by the ***anterior/ventral primary rami of upper four (4) cervical spinal nerves.***
- These rami/roots appear between the anterior and posterior tubercles of transverse processes of cervical vertebrae.
- These four roots connected to each other to form three (3) main loops.



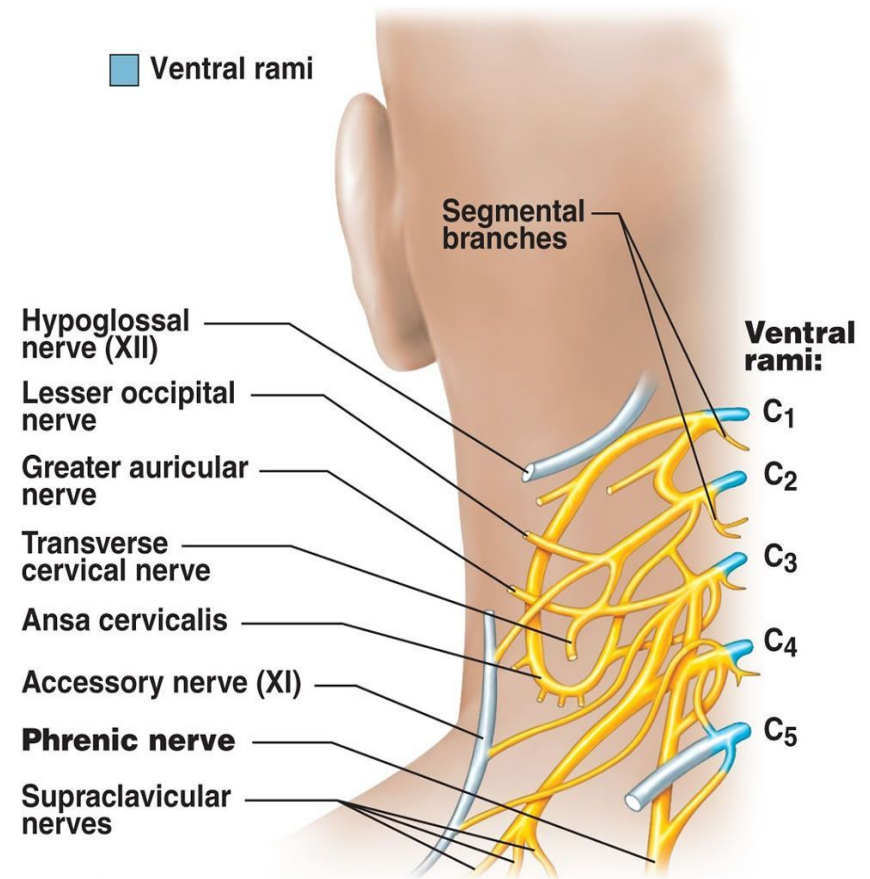
Relations of the cervical plexus:

- Posteriorly, it is related to the levator scapulae & scalenus medius muscles, and
- Anteriorly, to prevertebral fascia, the internal jugular vein, & the sternocleidomastoid muscle.



Branches of the Cervical Plexus:

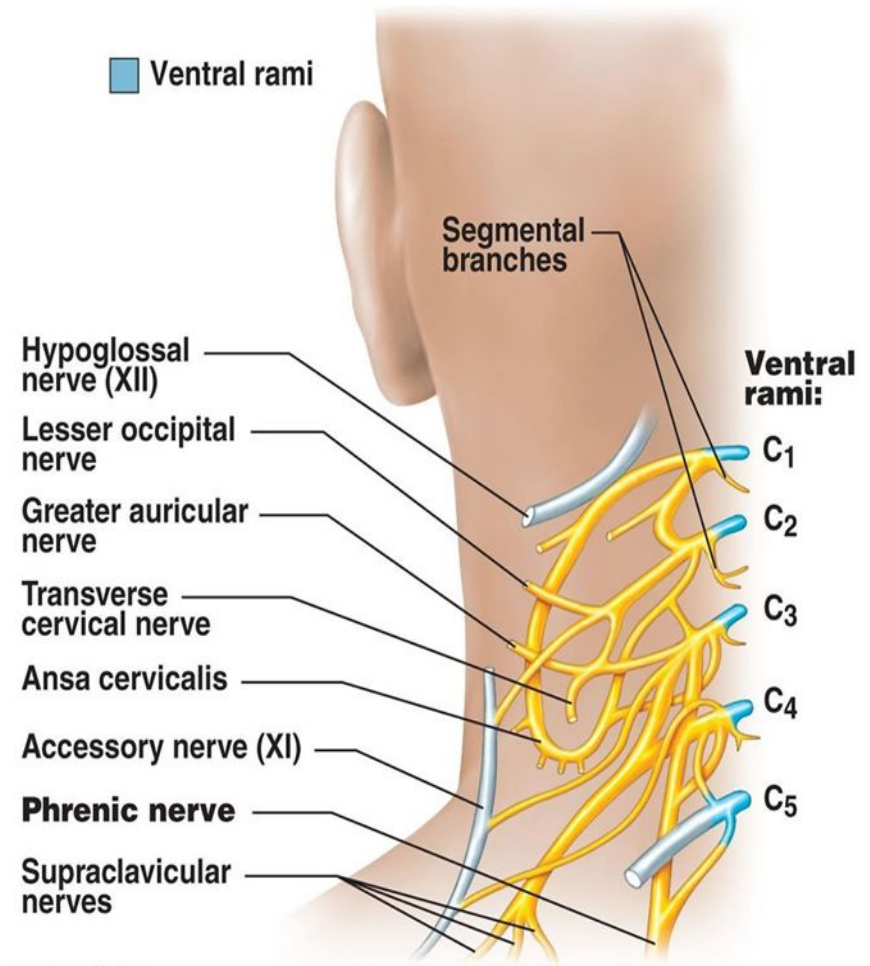
- **Superficial/cutaneous Branches:**
 - i. Lesser occipital (C2),
 - ii. Great auricular (C2, C3),
 - iii. Transverse/anterior cutaneous nerve of the neck (C2, C3),
 - iv. Supraclavicular (C3, C4)



• Deep Branches:

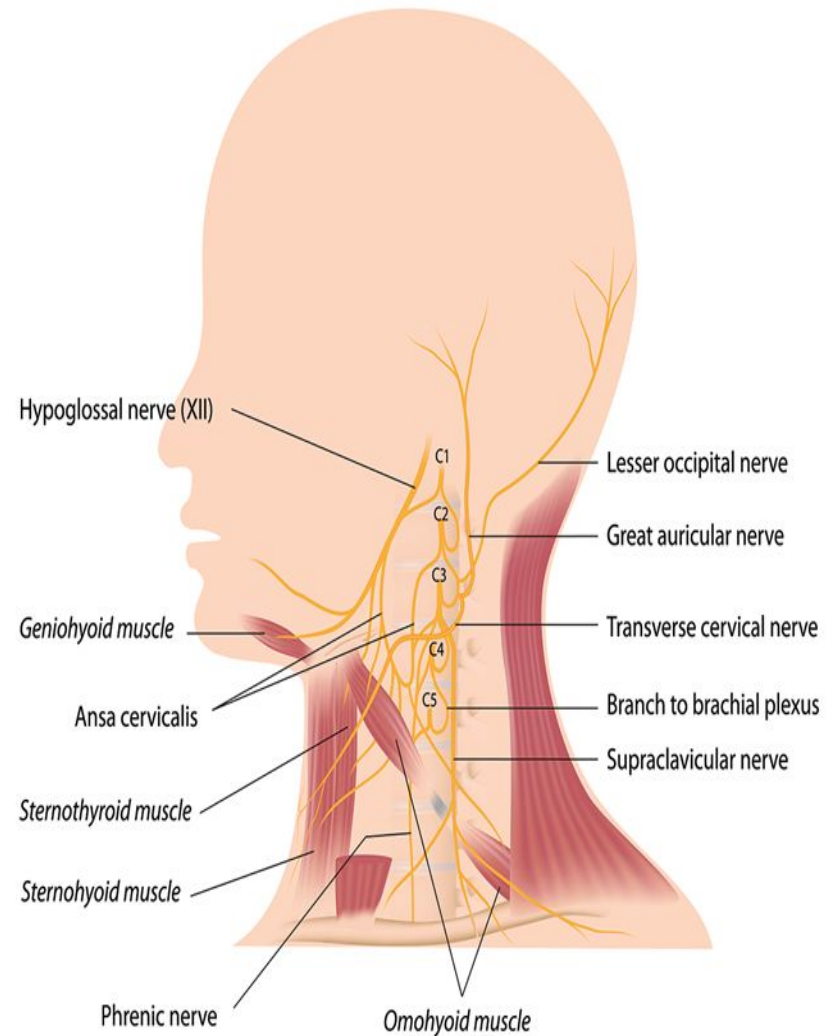
Communicating Branches:

- a) **Gray rami communicans** pass from the superior cervical ganglion to the roots of C1-C4 nerves.
- b) A **branch from C1 joins the hypoglossal nerve** and supplies the thyrohyoid and geniohyoid muscles and superior belly of omohyoid (through ansa cervicalis).
- c) A **branch from C2 to sternocleidomastoid and branches from C3 and C4 to trapezius join the accessory nerve**.

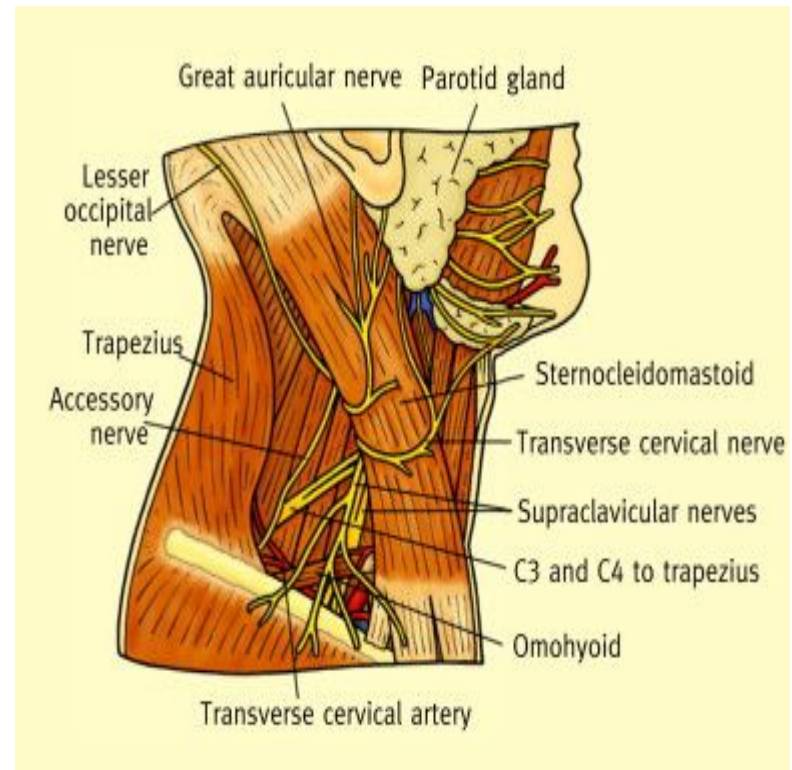


Muscular Branches

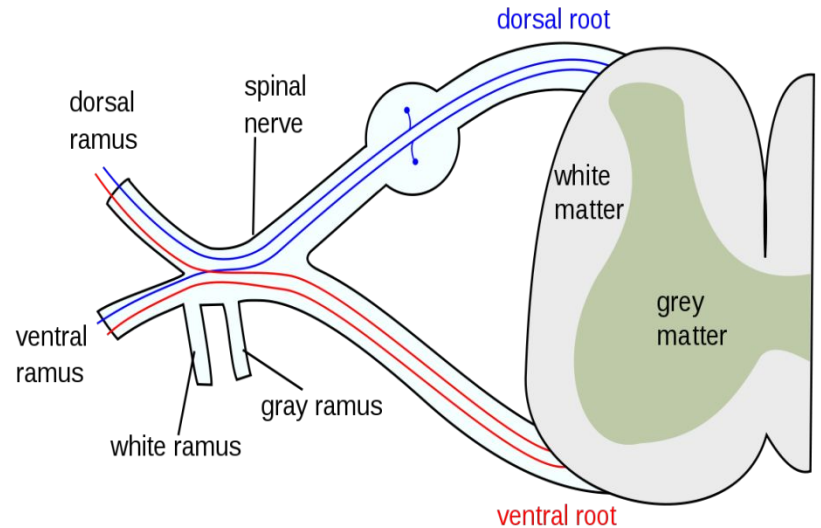
- The main muscle solely supplied by the cervical plexus:
- i. **Lower root of ansa cervicalis (descendens cervicalis)** to the infrahyoids (sternohyoid, sternothyroid & inferior belly of omohyoid).



- The main muscles supplied by cervical plexus along with brachial plexus or spinal accessory nerve:
 - i. **C2 along with accessory nerve** to sternocleidomastoid.
 - ii. **C3, C4 along with accessory nerve** to trapezius.
 - iii. **C3, C4 with C5 (dorsal scapular nerve)** to levator scapulae.
 - iv. **C3, C4, C5 (phrenic nerve)** to the diaphragm.

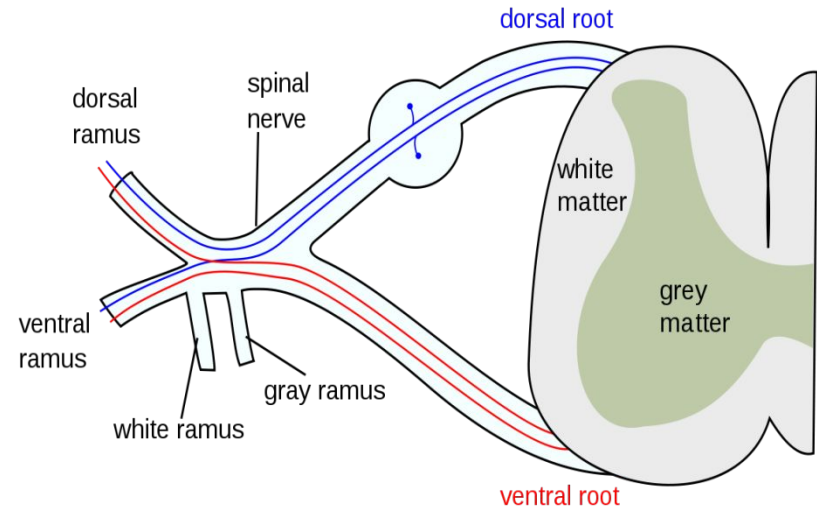


- It connects autonomic sympathetic chain with the anterior ramus of a spinal nerve.
- Present at every level of the spinal cord.
- They are mainly responsible for carrying the post-ganglionic nerve fibers to their destination.



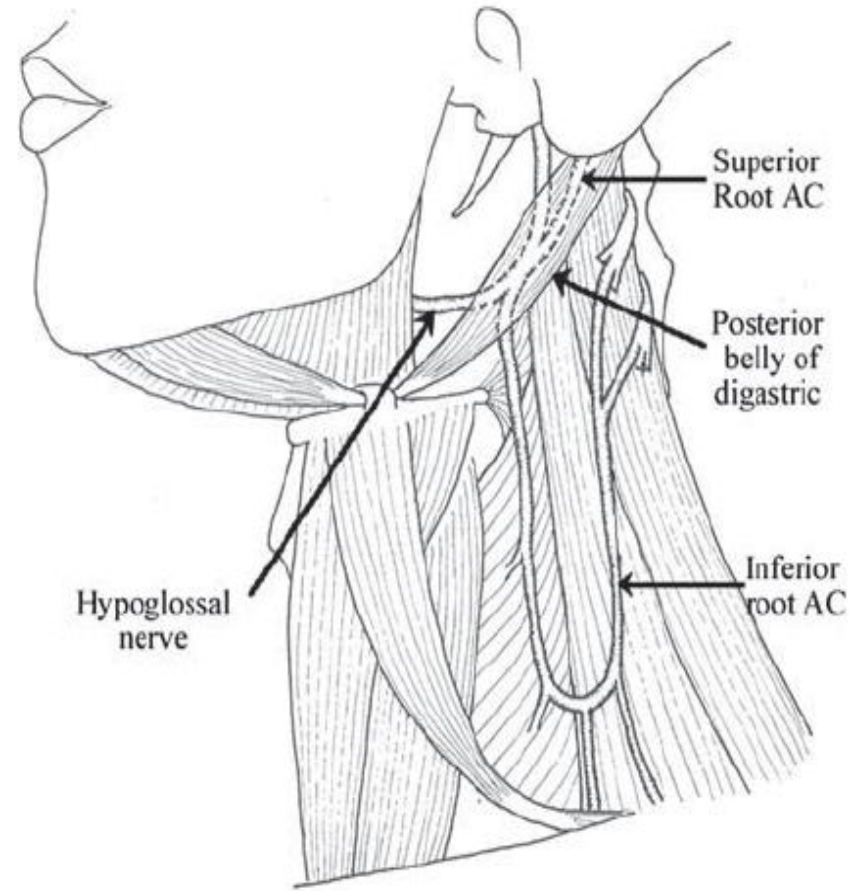
White rami communicans:

- It connects preganglionic sympathetic nerve fibers of the spinal cord to the sympathetic trunk.
- Found between spinal cord segments T1-L2.



Ansa Cervicalis/Ansa Hypoglossi:

- It is a thin nerve loop embedded in the anterior part of carotid sheath.
- It has a superior root derived from first cervical spinal nerve (C1) and joins the hypoglossal nerve.
- Its inferior root is derived from second and third cervical spinal nerves (C2, C3) and continues inferiorly to join superior root.



Distribution:

- Superior root to superior belly of omohyoid.
- Ansa cervicalis to sternohyoid, sternothyroid, and inferior belly of omohyoid.

